The eigenvalues are $\pm i\omega$. The reason a method like Euler's method can never perform well for molecular dynamics is that molecular dynamics is a Hamiltonian system and at the bottoms of basins on the energy surface, which correspond to stable centers, we expect all the eigenvalues of a local linearization of the problem to be purely imaginary. The stability condition always fails to hold and, for Euler's method, z_n grows exponentially rapidly away from the equilibrium point.

Let us perform a similar analysis for the Symplectic Euler method. Because Symplectic Euler cannot be applied to scalar equations (due to its partitioned structure) we must work directly with the second order system, but this is straightforward for the harmonic oscillator. The timestep map is defined by

$$\begin{aligned} Q &= q + hP, \\ P &= p - h\Omega^2 q. \end{aligned}$$

Solving for Q, P this yields

$$\begin{bmatrix} Q \\ P \end{bmatrix} = \begin{bmatrix} 1 - h^2\Omega^2 & h \\ -h\Omega^2 & 1 \end{bmatrix} \begin{bmatrix} q \\ p \end{bmatrix}.$$

The eigenvalues of the matrix are easily found, they are

$$\lambda_{1,2} = 1 - \frac{h^2\Omega^2}{2} \pm \frac{1}{2}\sqrt{h^4\Omega^4 - 4h^2\Omega^2}.$$

Then we observe that for $h^2\Omega^2 \leq 4$, they are complex and their squared magnitude is

$$|\lambda_{1,2}|^2 = \left(1 - \frac{h^2\Omega^2}{2}\right)^2 + \frac{1}{4}\left(4h^2\Omega^2 - h^4\Omega^4\right) = 1,$$

thus both eigenvalues lie on the unit circle in the complex plane as long as

$$h\Omega \leq 2.$$

This is the so-called linear stability condition of the Symplectic Euler method: if $h\Omega \leq 2$ the integrator is stable. When $h\Omega > 2$, the eigenvalues of the discretization method are both real, with one strictly inside and one strictly outside the unit circle. This implies that the method will exhibit exponentially growing solutions. We say that the *stability threshold* of the Symplectic Euler method is $2/\Omega$.

Among explicit symplectic Partitioned Runge-Kutta methods this is the maximum stability threshold [74]. In a similar way one can analyze the stability of the Verlet and other methods and one thus obtains conditions on the stepsize that must hold for the equilibrium points to be stable in the linearization. Analyzing the stability of both continuous and discrete iteration is much more complicated for